

CHEMISTRY

Robust, scalable, and highly selective spirocyclic catalysts for industrial hydroformylation and isomerization-hydroformylation

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Hydroformylation (HF) or isomerization-hydroformylation (ISO-HF) represents the most direct and practical route for producing aldehydes on an industrial scale. To resolve the issues of low activity, low linear/branched (*l/b*) ratio, and low stability in HF and ISO-HF, we herein reported a class of spirocyclic diphosphites. Notably, the ligand termed O-SDPhite afforded excellent catalytic activity and regioselectivity for the HF of various olefins. Excellent *l/b* ratio and an unprecedented turnover number of up to 17,620,000 were achieved. O-SDPhite was also found to be effective in the regioselective ISO-HF of the industrially related cheap and abundant C4 *Raffinates* to *n*-valeraldehyde produced on a multimillion-ton scale. The reaction with O-SDPhite, superior to that of benchmark Biphephos, was continuously operated for 41 days and afforded an average 38.6 *l/b* ratio (31 days and 14.7 *l/b* ratio for Biphephos).

INTRODUCTION

Hydroformylation (or oxo synthesis) is one of the most important and largest homogeneous catalytic reaction in industry. Since its discovery by Otto Roelen in 1938 (1), several well-known commercialized processes such as the BASF-Oxo process (2), Shell process (3), LP Oxo process (4, 5), and Ruhrchemie/Rhône-Poulenc (RCH/RP) process have been developed. It is estimated that more than 10 million tons of aldehydes are formed annually via such transformation (6, 7), of which more than 6 million tons are for the phthalate plasticizer industry (8, 9) with a revenue of around \$10 billion (Fig. 1A) (10, 11). Besides the aldol route for phthalate plasticizers, the aldehyde oxidation or oxidation-esterification products are utilized intensively in synthesizing polyol esters (POEs) through Fischer esterification or transesterification with polyols (Fig. 1A) (12). POEs are extensively applied in automotive and refrigerant lubricants, cosmetics, etc. (13, 14), with a market value of around \$5 billion (15). To attain high activity and regioselectivity, commercial hydroformylation of propylene or *Raffinate II/III* uses unmodified/modified Co or Rh catalysts (16–18). Despite the great advances in cocatalyzed hydroformylation by Stanley *et al.* (19, 20) and Zhang *et al.* (21), Rh catalyst still excels Co catalyst by 20 to 1000 times in activity. Rh catalysts are operated at 70 to 120°C and 10 to 20 bar, whereas >140°C and >30 bar are needed for Co catalysts.

Phosphorus ligands play a pivotal role in the development of hydroformylation. For instance, a plethora of monodentate, bidentate, and multidentate ligands have been developed for this purpose, including TPP (triphenylphosphine) (22); Bisbi (23–26); Xantphos (27–29); Biphephos (30, 31); Naphos (32, 33) (Fig. 1B); pyrrole-based bis-phosphoramidites (34–36), tri-phosphoramidites (37),

and tetra-phosphoramidites (38); spiroketal-based phosphorus ligands (39, 40); and arylphosphites (41, 42), and have been shown to exhibit good regioselectivities in the hydroformylation of both terminal and internal olefins. In particular, bidentate/multidentate phosphine, phosphite, or phosphoramidite ligands, which have a natural bite angle (β_n) of about 110°, will favor the linear product (43, 44). From the perspective of reactivity and selectivity, the most successful ligand types are Bisbi (25), Xantphos (27), and Biphephos (45) (Fig. 1B). Electron-deficient phosphite ligands, such as the state-of-the-art Biphephos, showed much higher reactivity toward the hydroformylation. Besides developing phosphorus ligands, dual or bimetallic catalysis (46, 47), biphasic catalysis (48), micellar catalysis (49), or single-atom catalysis (50) was also introduced for Rh-catalyzed HF. Despite these attempts, from an industrial perspective, relatively few have been utilized in the scaling-up process (i.e., from bench to pilot scale); the most highly active ligands and Rh catalysts reported in the literature stay at laboratory scale. The bidentate ligands like BISBI, Xantphos, and Naphos are not used in commercial hydroformylation, probably due to Rh-induced phosphine ligand fragmentation problems. The bisphosphite Biphephos also has the tendency for degradation by insertion of Rh to the C–C bond of the biphenyl moiety (51). Industrial applied hydroformylation still utilizes unmodified Co or mono- and diphosphine modified Rh catalysts, which were invented 35 to 80 years ago (TPP/TPPTS, Bisbi, Xantphos, and Biphephos). Hence, the development of highly effective, chemoselective, regioselective, and stable catalytic systems that will be suitable for both terminal and internal olefins still remain challenging and indispensable.

Because of their high rigidity and stability, chiral ligands based on privileged spiro scaffolds, such as SPINOL (52–58) and spiroketal (59–61), have demonstrated excellent performance in asymmetric catalysis; nevertheless, they were rarely explored in olefin hydroformylation. Our group has previously reported structurally unique diphenol O-SPINOL and related O-SpiroPAP (62), O-SDP (63), and O-SIPHOX (64) ligands for asymmetric catalysis. The introduction of oxygen atoms greatly increased the distance between the two phenolic oxygens in O-SPINOL, resulting in a larger bite

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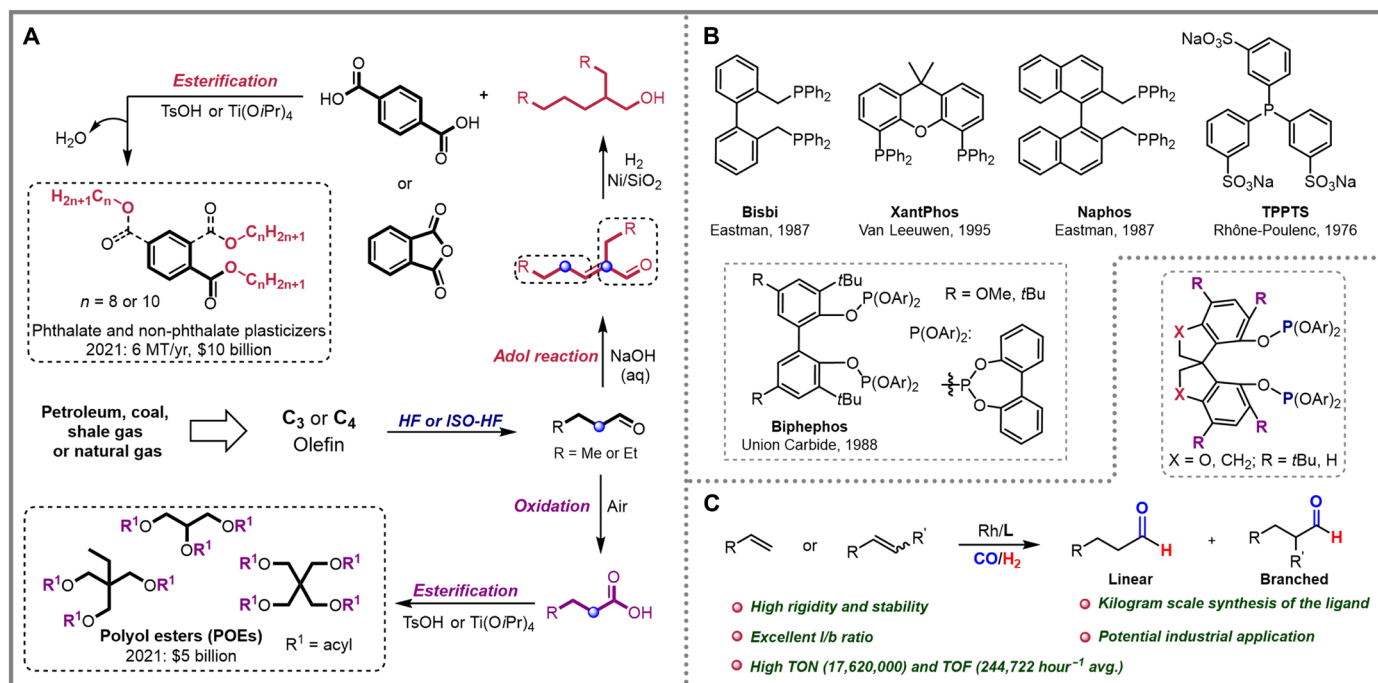


Fig. 1. Industrial applications of HF or ISO-HF and representative diphosphine ligands for industrial HF. (A) Industrial application for linear-selective HF or ISO-HF. **(B)** Selected examples of diphosphine ligands utilized in industrial HF or ISO-HF. **(C)** Spirocyclic diphosphite ligands for linear-selective HF or ISO-HF.

angle of the resulting ligands (62, 63). According to our Density Functional Theory (DFT) calculations (for details, see section S12.1), the bite angle data for ligands **L27** and **L28** were 111.8° and 110.9°, respectively, which is consistent with our assumption that the introduction of oxygen atoms will result in a larger bite angle. Thus, we envision that the diphosphite or diphosphoramidite analog of O-SPINOL will lead to a much bigger bite angle and a higher linear/branched (*l/b*) ratio under hydroformylation conditions. On the basis of the successful example of Biphephos and tetrakisphosphite-*t*Bu (TBTP) ligands (65), introducing sterically hindered *tert*-butyl groups in the *ortho*-position of O-SPINOL will further improve the speed (66), selectivity (16), and stability of the ligand (67) (Fig. 1C).

RESULTS

Ligand evaluation

By Friedel-Crafts alkylation of O-SPINOL (65) with *t*BuOH, *t*Bu-O-SPINOL was obtained in 99% yield. Reaction of 1,1'-biphenyl-2,2'-dioxychlorophosphine with *t*Bu-O-SPINOL afforded the corresponding *oxa*-spirocyclic diphosphite ligands **L27**, termed O-SDPhite, in 76% yield on the kilogram scale after recrystallization (for details, see fig. S1). **L28** (SDPhite), **L29**, and **L30** were prepared similarly. With the synthesized ligands in hand, we examined their performance in the Rh-catalyzed hydroformylation with 1-octene as model substrate, and the results are summarized in Fig. 2. Ligands **L27** and **L28** outperformed various mono-, bi-, and multidentate phosphine/phosphoramidite/phosphite ligands of varied steric and electronic properties in reactivity. Notably, the best *l/b* ratio of 96.4 was achieved with *oxa*-spirocyclic ligand **L27**. In contrast, the SPINOL-based analog **L28** could only give an *l/b* ratio of 39.9. As expected, 0.1 to 7.3% isomerized C₈ alkenes were observed for all monodentate ligands

(**L1** to **L6** and **L8**) because the regioselectivity was poor with both linear and branched products formed at almost the same rate (Fig. 2), whereas for the xanthene, biaryl, and spiro ligands [**L15**, **L16** to **L22** (68), **L27**, **L28**, and **L30**], isomerization can occur before or simultaneous with hydroformylation, leading to increased isomerized alkene of 7.3%, 15.3 to 32.8%, and 15.4 to 27.4%, respectively. Compared to benchmark ligands Xantphos (**L15**), Bisbi (**L16**), and Biphephos (**L20** and **L21**), conversion with **L27** was improved by 12 and 10% (99.6% conversion compared with 87.6% of **L15** and 89.6% of **L17**), the *l/b* ratio was 2.3, 2.0, 1.4, and 3.6 times higher than that of **L15**, **L16**, **L20**, and **L21**, and the yield of linear aldehyde was improved by 4.9%, 5.6%, 15.6%, and 14.2%, respectively (Fig. 2).

Scope of the reaction

L27 turned out to be the best ligand in linear hydroformylation of 1-octene (**1a**). Optimization experiments were conducted subsequently. After thorough screening, the optimal reaction condition was identified as 0.05 mol% catalyst loading, *L*/Rh = 4, H₂/CO (1:1, 14 bar), 100°C, 2 hours, and toluene as solvent (see table S3 for details). Given the above condition, we next examined the generality and limitations of **L27** in the hydroformylation of different alkene substrates. As shown in Fig. 3, aliphatic alkenes gave the corresponding aldehyde products in high conversions with excellent chemo- and regioselectivities. Among these substrates, 1-hexene (**1b**) and 1-heptene (**1c**) afforded the corresponding aldehydes (**2b** and **c**) with *l/b* ratios of 96 and 99, respectively (Fig. 3, entries 1 and 2). Abated with low *n*-selectivity, hydroformylation of 1-decene (**1d**) gave nearly full conversion (>99%). The eroded *l/b* ratio was due to the reactivity and thermodynamic properties of the long-chain alkenes because the double bond at the terminal position always favored the inward migration (Fig. 3, entry 3). Gratifyingly, our Rh/**L27** system

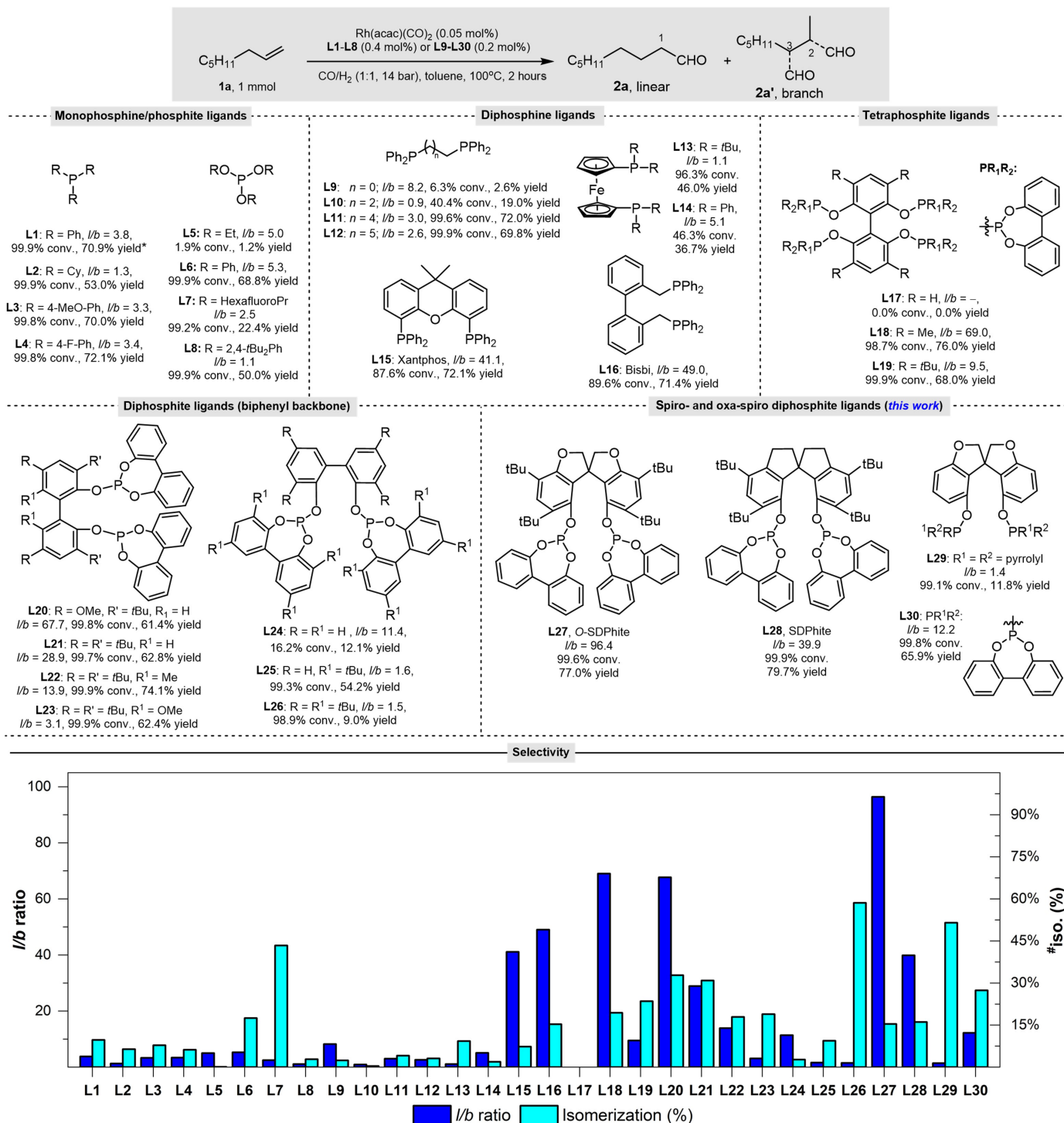


Fig. 2. Rh-catalyzed hydroformylation of 1-octene using various ligands. Reaction conditions: The reaction was conducted in a 50-ml autoclave, with alkene (1.0 mmol), Rh(acac)(CO)₂ (0.05 mol%), **L1 to L8** (5.0 mol%), **L9 to L30** (0.2 mol%), H₂/CO (1:1, 14 bar), and toluene (0.5 ml), and stirred at 100°C for 2 hours. *Yields are given for the linear aldehyde; #iso. (%) refers to the percentage of isomerized olefins including 2-, 3-, and 4-octene as well as their cis/trans isomers.

Entry	Alkene	Linear aldehyde	Conv./Sel. (l/b)/yield [%]	Entry	Alkene	Linear aldehyde	Conv./Sel. (l/b)/yield [%]
1			>99/ 99 (99)/ 62	11*			>99/ 99 (244)/ 97
2			>99/ 99 (96)/ 77	12			96/ 79 (3.7)/ 52
3			>99/ 78 (3.5)/ 68	13			>99/ 81 (4.2)/ 72
4			91/ 95 (21)/ 74	14			>99/ 99 (80)/ 75
5			92/ 95 (18)/ 62	15			>99/ 95 (25)/ 72
6			>99/ -- / 73	16			>99/ 87 (6.8)/ 82
7			>99/ -- / 94	17			>99/ 81 (4.2)/ 54
8			>99/ 99 (131)/ 87	18			>99/ 99 (69)/ 70
9			90/ 93 (16)/ 39	19			>99/ 81 (4.2)/ 35
10			>99/ 99 (91)/ 94	20			97/ 81 (4.3)/ 57

Fig. 3. Rh-catalyzed HF or ISO-HF of aliphatic, aromatic, and functionalized alkenes. Reaction conditions: The reaction was conducted in a 50-ml autoclave, with alkene (1.0 mmol), Rh (acac)(CO)₂ (0.05 mol%), **L27** (0.2 mol%), H₂/CO (1:1, 14 bar), and toluene (0.5 ml), and stirred at 100°C for 2 hours. For **1e** and **1f**, 0.5 mmol, 0.25 mol% catalyst loading, H₂/CO (1:1, 20 bar), toluene (1.0 ml), 125°C, 4 hours and 2 hours, respectively; for 1 hour, *T* = 140°C; for **1t**, H₂/CO (3:1, 14 bar). #Sel. (%) refers to the ratio of linear aldehyde compared with the sum of linear and branched aldehyde. *When the reaction of **1l** (6.0 mmol) in 0.12 ml of toluene was conducted at 110°C with S/C (substrate-to-catalyst ratio) = 20,000,000, 88.1% conv., 86.4% yield, and >99% linear selectivity were achieved after 72 hours.

demonstrated excellent isomerization-hydroformylation (ISO-HF) ability for internal alkenes such as 2-hexenes (**1e**) and 2-octenes (**3f**), which gave >90% conversions and ≥95% linear selectivities (Fig. 3, entries 4 and 5). Cyclic olefins such as norbornene (**1g**) and cyclohexene (**1h**) were converted >99% into the desired aldehydes (Fig. 3, entries 6 and 7). For 4-methyl-1-pentene (**1i**), >99% conversion and >99% linear selectivity were achieved, whereas 2-methyl-1-pentene (**1j**) gave eroded conversion and linear selectivity (90 and 93%). Next, a variety of aromatic alkenes and functionalized alkenes were tested. For example, styrene (**1t**) and allyl phenyl ether (**1u**) showed excellent reactivities, with moderate linear selectivities (Fig. 3, entries 19 and 20). The tolerance of different functionalized olefins bearing with silyl, acetal, ester, and halogen

groups (**1k** to **1s**) is important from a synthetic perspective. As shown in entries 10 and 11 in Fig. 3, the ethylene derivatives with different silyl groups (**1k** and **1l**) exhibited excellent reactivities, aldehyde yields, and linear selectivities. Notably, the scaling up experiment of **1l** with a substrate-to-catalyst ratio of 20,000,000 could give 88.1% conversion and 84.6% yield with retention of regioselectivity [turnover number (TON) = 17,620,000, average TOF = 244,722 hour⁻¹]. Vinyl acetals (**1m** and **1n**) were fully converted into desired aldehydes in high conversions (96 to 99%) and moderate linear selectivities (79 to 81%). Notably, condensation of 4,4-diethoxy-1-butanal (**2n**) with dimethylamine could give 4,4-diethoxy-*N,N*-dimethyl-1-butanamine, which is an important pharmaceutical intermediate for antimigraine drugs such

as sumatriptan and rizatriptan. Moreover, 5-hexenyl acetate (**1o**) and methyl pentanoate (**1p**) were fully converted into desired linear aldehydes with high *l/b* ratios (25 to 80), whereas 4-pentenyl acetate (**1q**) gave eroded linear selectivity (Fig. 3, entries 14 to 16). Methyl acrylate (**1r**) displayed similar linear selectivity to styrene (**1t**) due to the electronic effect. Alkenyl halide (**1s**) was also evaluated, affording high *l/b* ratio (67) and full conversion (Fig. 3, entry 18).

After demonstrating the viability of the Rh/**L27** catalyst for various olefins, we next examined the performance of our system with 1-octene at low catalyst loadings (for details, see table S4). Gratifyingly, the conversion and regioselectivity were maintained at a substrate-to-catalyst ratio of 10,000,000; this is remarkable from an industrial viewpoint because the Rh catalyst loading can be decreased to as low as 0.1 parts per million (ppm). To the best of our knowledge, this is also a rare example that the hydroformylation of long-chain alkene proceeded at ultralow catalyst loading with TON = 9,720,000 and an average TOF value of 135,000 hour⁻¹. The effect of ligand amount was also investigated at a substrate-to-catalyst ratio of 1,000,000 (table S4, entries 6 and 7), and there were no discernible decreases in the aldehyde yield or linear selectivity when the ligand amount was decreased to 0.001 mol% (**L27**/Rh = 10). The instantaneous TOF, which reflected the intrinsic activity of Rh/**L27** catalyst, was calculated by conversion diagram, affording an excellent TOF value of 952,800 hour⁻¹ (for details, see table S6).

HF or ISO-HF of propylene and butenes

As previously indicated in Fig. 1A, hydroformylation of propylene or mixed butenes is a crucial step in Dow and Johnson Matthey's LP OxoSM technology for the production of *n*-butanal and *n*-pentanal (4). Mixed butenes are cheap and abundant and can be easily accessed from methanol to olefin plants, Fischer-Tropsch plants, and ethylene oligomerization (16). The ISO-HF of mixed butenes is challenging because the reactivity of (*Z*)-butene and (*E*)-butene was very low, and isomerization was necessary before hydroformylation for the production of linear aldehyde (69). The HF of propylene was first investigated by comparing the benchmark ligand Biphephos (**L21**) with O-SDPhite (**L27**). To our delight, an *l/b* ratio of 40.8 and 88.9% yield were obtained, which are much better than those of the benchmark ligand Biphephos (Fig. 4, entry 1). Next, butene and mixed butenes were examined under industrially relevant conditions in a 150-ml semi-batch reactor (for details, see section S3.11). Pure (*Z*)- and (*E*)-2-butenes were used individually as feed (Fig. 4, entries 2 and 3). As expected, our system performed well and **L27** surpassed **L21**. The benchmark ligand **L21** could only afford *l/b* ratios of 13.6 and 19.9, respectively. In sharp contrast, *l/b* ratios of 43.2 and 47.2 were obtained with **L27**. In addition, the SPINOL-based diphosphite **L28** was also used for comparison, though it only provided (*Z*)- and (*E*)-butenes with *l/b* ratios of 17.4 and 21.5, respectively. When using various mixed butenes as feed, i.e., *Raffinate II*, *Raffinate III*, and etherified C4 in ISO-HF, again, our catalytic system proved to be superior compared to the present state-of-the-art industrial catalysts. The *l/b* ratios for **L27** (45.7 to 51.7) were almost twice as much as that of **L21** (25.8 to 30.6), and the yields of linear aldehyde were improved by 0.6 to 5.3%; notably, up to 80% yield of **2w** was achieved for *Raffinate III* (Fig. 4, entry 4). This is convenient for consecutive aldol condensation to produce 2-propyl-2-heptenal, because **L27** almost inhibits the formation of unwanted and valueless isovaleraldehyde, which is beneficial for the purification and saving cost.

To give a full insight into the origin of high reactivity and linear selectivity of our catalytic system (Rh/O-SDPhite), DFT calculations were conducted with 1-butene as model substrate [calculations were performed by means of the Gaussian 16 suite of programs (70)]. DFT calculations were conducted using BP86-D3 functional. The Stuttgart–Dresden (SDD) basis set and Electrochemical Potencial (ECP) were used to describe Rh (71). The 6-31G(d) basis set was used for all remaining atoms (C, H, P, and O). Frequency calculations were carried out for optimized structures to verify the stationary points or transition states at the same level. Polarization functions ($\zeta_f = 1.350$) were added for Rh (72). The solvent effect of toluene was considered using the Solvent Model based on Density (SMD) model (73). Geometric structures of all species were optimized at $T = 373.15$ K and 10.0 atm (for details, see section S12). **Int1A** with equatorial-equatorial configuration is lower in energy by 7.35 kcal/mol compared to **Int1B** with an equatorial-apical configuration, indicating that **Int1B**, which is beneficial for the formation of branched product, can be omitted, whereas a much smaller energy difference of 3.33 kcal/mol was calculated for Biphephos (Fig. 5A). In addition, the free energies of transition states leading to linear and branched products (**TS1A**, **TS1B**, **B-TS1A**, and **B-TS1B**) were also calculated, providing an energy difference of 2.75 kcal/mol for O-SDPhite and 2.22 kcal/mol for Biphephos, respectively (Fig. 5B). These results were in accordance with our experimental observations.

ISO-HF of *Raffinate III* at semi-batch scale

After demonstrating the distinct advantages of Rh/**L27** over Rh/**L21** catalysts at the semi-batch scale, we performed an industrial case study in three continuous stirred tank reactors (CSTRs) in a serial system (5.0 liters each) (Fig. 6B; for details, see section S3.12). Again, the benchmark **L21** was compared with **L27** in the ISO-HF of *Raffinate III*. As depicted in Fig. 6A, in a continuous operation of 41 days (984 hours) under steady-state conditions, the Rh/**L27** system showed unprecedentedly high regioselectivity (average *l/b* ratio of 38.3) and 72.3% average yield of *n*-pentanal, affording 404 kg of *n*-pentanal in total. In sharp contrast, **L21** gave an eroded *l/b* ratio of 14.7, providing only 282 kg *n*-pentanal and an average aldehyde yield of 68.5% over a period of 31 days (744 hours), indicating that our Rh/**L27** system was much more robust, endurable, and regioselective. The hydroformylation using a flow reactor was conducted with *n*-pentanal (99.9% purity) as solvent initially, which could account for the drop of *l/b* ratio in the first few days. About 5 days later, the fluctuation in *l/b* ratio decreased. The marked drop of *l/b* ratio in the last few days was due to the decomposition of the ligand, and the formation of less selective monophosphite ligand will result in the reduction in regioselectivity (for details, see section S8). Evidently, our catalytic system demonstrated superior performance compared to the state-of-the-art industrial catalyst Rh/**L21** in the ISO-HF of mixed butenes during continuous operation, resulting in substantial cost savings (for details, see section S9).

DISCUSSION

In conclusion, we have developed a highly reactive and effective catalytic system based on spirocyclic diphosphite ligand, termed O-SDPhite, and this ligand was applied in the HF and ISO-HF of feedstock olefins. It should be noted that excellent regioselectivities and yields were obtained for the ISO-HF of mixed butenes (*Raffinates*), indicating that O-SDPhite has high potential applications in the

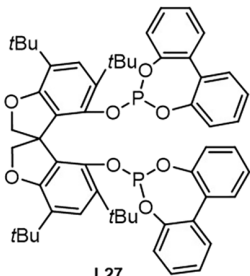
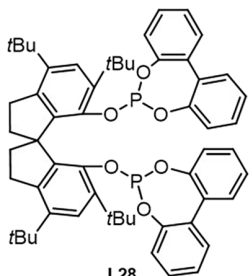
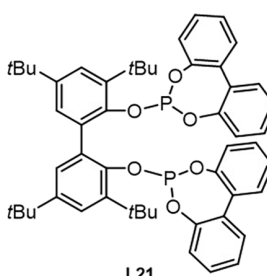
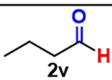
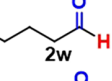
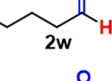
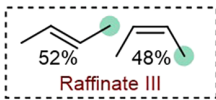
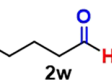
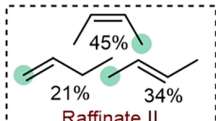
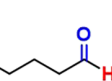
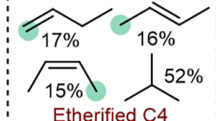
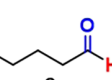
 L27  L28  L21										
Entry	Substrate	Product	Linear sel. (l/b)		Conv. (%)		Iso.# (%)		2v or 2w (%)	
			L27 (L28)	L21	L27 (L28)	L21	L27 (L28)	L21	L27 (L28)	L21
1	 Propylene	 2v	40.8	12.6	Full		--		88.9	74.3
2	 (Z)-butene	 2w	43.2 (17.4)	13.6	95.0 (90.9)	94.2	17.7 (16.7)	14.8	76.4 (71.6)	74.8
3	 (E)-butene	 2w	47.2 (21.5)	19.9	84.7 (81.2)	84.1	10.8 (8.5)	7.6	74.0 (71.0)	74.0
4	 Raffinate III	 2w	45.7	26.0	82.3	79.4	n.d.		80.0	75.7
5	 Raffinate II	 2w	47.2	25.8	72.3	72.9	n.d.		70.8	70.2
6*	 Etherified C4	 2w	51.7	30.6	78.4	73.9	n.d.		76.9	71.6

Fig. 4. HF or ISO-HF of propylene and butenes catalyzed by Rh/L27 and Rh/L21. Reaction conditions: For propylene, propylene (13.1 mmol, 6.5 bar in a 50-ml autoclave), Rh(acac)(CO)₂ (2.17 μmol, 51.2 ppm), L/Rh = 3, toluene (5.0 ml), CO/H₂ (1:1, 13.5 bar), 70°C, 50 min for L21, 45 min for L27. For (E)-butene, (Z)-butene, or Raffinates (40 mmol), Rh(acac)(CO)₂ (10 μmol, 159 ppm), L/Rh = 3, toluene (4.3 ml), CO/H₂ (1:1, 10 bar), 80°C, 4 hours, decane as the internal standard. #Iso. (%) refers to the percentage of isomerized olefins. *Etherified C4 (80 mmol), Rh(acac)(CO)₂ = 13.6 μmol.

chemical industry. The origin of high regioselectivity was elucidated by DFT calculations. In addition, an unprecedented 18 million TON was afforded, and an industrial case study demonstrated exceptional long-term stability in a continuous ISO-HF test, in which Rh/L27 produced 43.2% more *n*-pentanal than Rh/L21. Our catalytic system opens doors for highly efficient and selective C4 olefin utilization in petroleum and coal industries. The insights into the outstanding performance of the Rh/L27 catalyst and the DFT results will motivate further interests from both academia and industry.

MATERIALS AND METHODS

General experimental procedures

Unless otherwise mentioned, all experiments were carried out under Ar atmosphere in a glovebox or using standard Schlenk techniques. Commercially available reagents were used without further purification. NMR spectra were recorded on a Bruker Avance 600 spectrometer or Bruker Avance 600 spectrometer with tetramethylsilane as

internal standard. All new products were further characterized by High Resolution Mass Spectroscopy (HRMS). A positive ion mass spectrum of sample was acquired on a Thermo LTQ-FT mass spectrometer with an electrospray ionization source. Crystal structure was measured with a BRUKER APEX III diffractometer. Gas Chromatography (GS) analyses were measured on an Agilent 6890N or an Agilent 7820A system.

General procedure for substrate scope investigation

To an 8.0-ml vial was added the catalyst precursor Rh(acac)(CO)₂ (2.58 mg, 1.0 × 10⁻² mmol), L27 (36.4 mg, 4 × 10⁻² mmol), and anhydrous toluene (3.0 ml) in the argon-filled glovebox. The mixture was stirred for 10 min at room temperature to give a pale yellow solution.

In a glovebox, a glass vial with a magnetic stirring bar was charged with substrate (1.0 mmol), followed by the addition of the above-prepared Rh/L27 solution (150 μl, 5.0 × 10⁻⁴ mmol), and then decane (50 μl) was added as the internal standard. Finally, anhydrous toluene was added to a total volume of 0.5 ml.

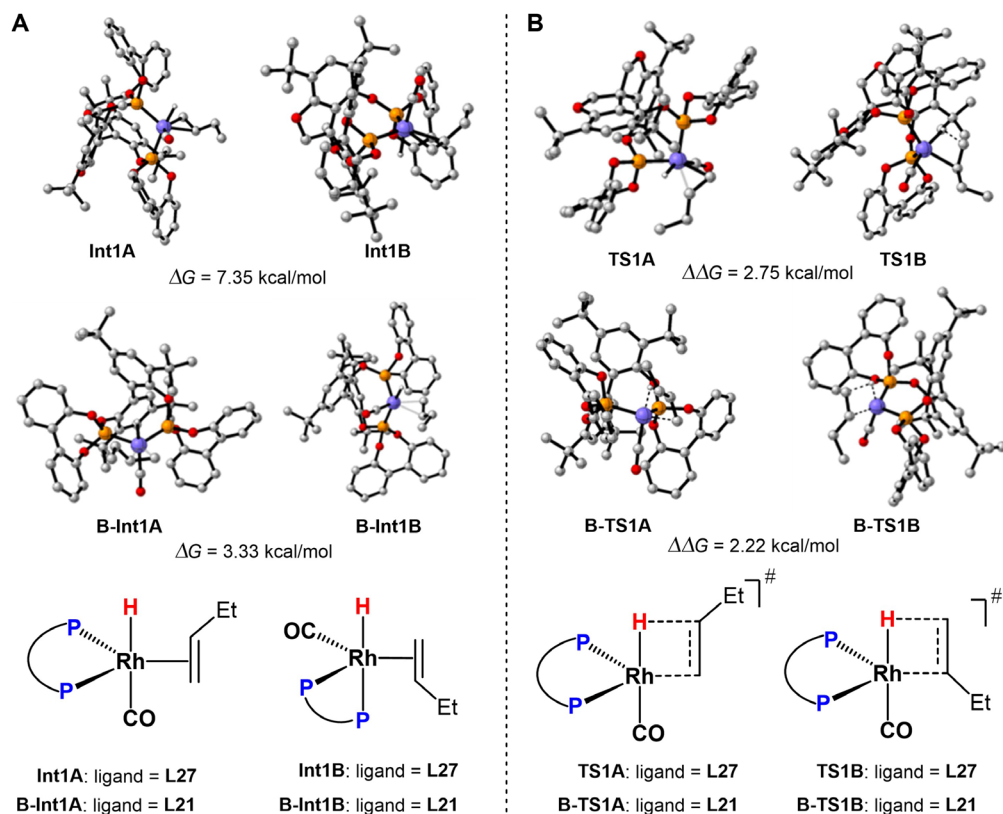


Fig. 5. Elucidation of origin of high linear selectivity by DFT calculations. (A) Optimized structures of intermediates with equatorial-equatorial and equatorial-apical configuration. (B) Structures of the transition states leading to linear and branched products.

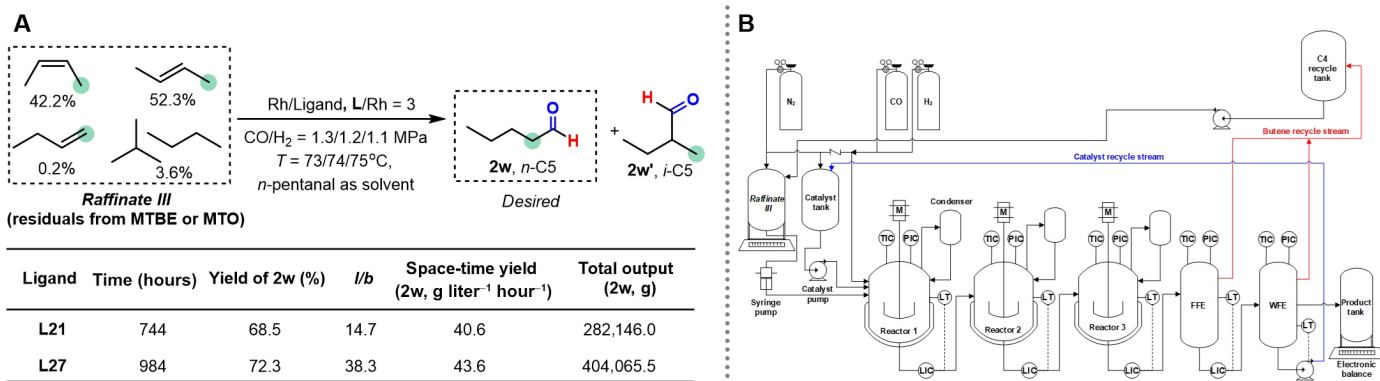


Fig. 6. Continuous-flow ISO-HF of Raffinate III in CSTRs. (A) Results for continuous-flow ISO-HF of Raffinate III. Reaction conditions: Rh(acac)(CO)₂ (38.4 mmol), L/Rh = 3, *n*-pentanal (12 kg, 15 liters) as solvent, Rh concentration in the catalyst tank = 323 ppm, catalyst feed rate = 15.5 ml/min, reactor 1/2/3 pressure = 1.3/1.2/1.1 MPa (H₂/CO = 1:1), reactor 1/2/3 temperature = 73°/74°/75°C, liquid hold-up for reactor 1/2/3 = 3.25/3.25/3.25 liters, residence time (liquid phase) for reactor 1/2/3 = 2.0/2.0/2.0 hours. (B) Process flow diagram for ISO-HF of C4 Raffinate. TIC, PIC, and LIC: temperature indicator and controller, pressure indicator and controller, and level indicator and controller; LT: level transmitter; FFE: falling-film evaporator; WFE: wiped-film evaporator.

The vial was then transferred to a 50-ml autoclave, which was then sealed and flushed with syngas CO/H₂ (1/1) three times and charged to 14 bar. The autoclave was then transferred to a pre-heated oil bath and stirred at 100°C for 2 hours before being cooled to room temperature. The reaction mixture was analyzed by GC to determine the conversion, percentage of aldehyde, and regioselectivity (*l/b* ratio).

Supplementary Materials

This PDF file includes:

Supplemental Materials and Methods
Figs. S1 to S40
Tables S1 to S14
DFT calculations
NMR and GC spectra
References

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